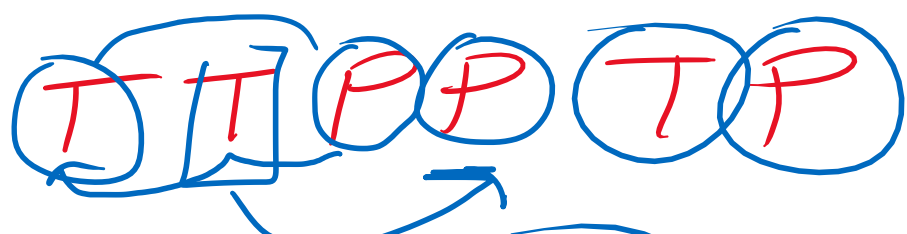


* Policeman Catch Thieves

P T T P T $k=1, n=5$

$$T = 1 + 1 = 2$$

Maximize



$$T = 1 + 1 + 1 = 3$$

Array: $\rightarrow$ $\begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 2 & 3 & 4 & 5 & 6 \end{bmatrix}$

P-1

T-0

* Police = $\begin{bmatrix} \cancel{0}, \cancel{1}, \cancel{2}, \cancel{3}, 4, 5, 6 \end{bmatrix}$ ✓ Thieves

* Thief = $\begin{bmatrix} \cancel{0}, \cancel{1}, \cancel{2}, \cancel{3}, \cancel{4}, \cancel{5}, \cancel{6} \end{bmatrix}$ ✓ Count

[Match $k=1$] F T - empty 1, 2, 3

(queue) (degree) $\begin{matrix} P & \cancel{1} & \cancel{2} & 3 & 4 & 5 \\ T & 3 & 5 & & & \end{matrix}$

$\begin{matrix} P & 3 & 4 \\ T & \cancel{2} & 5 \end{matrix}$

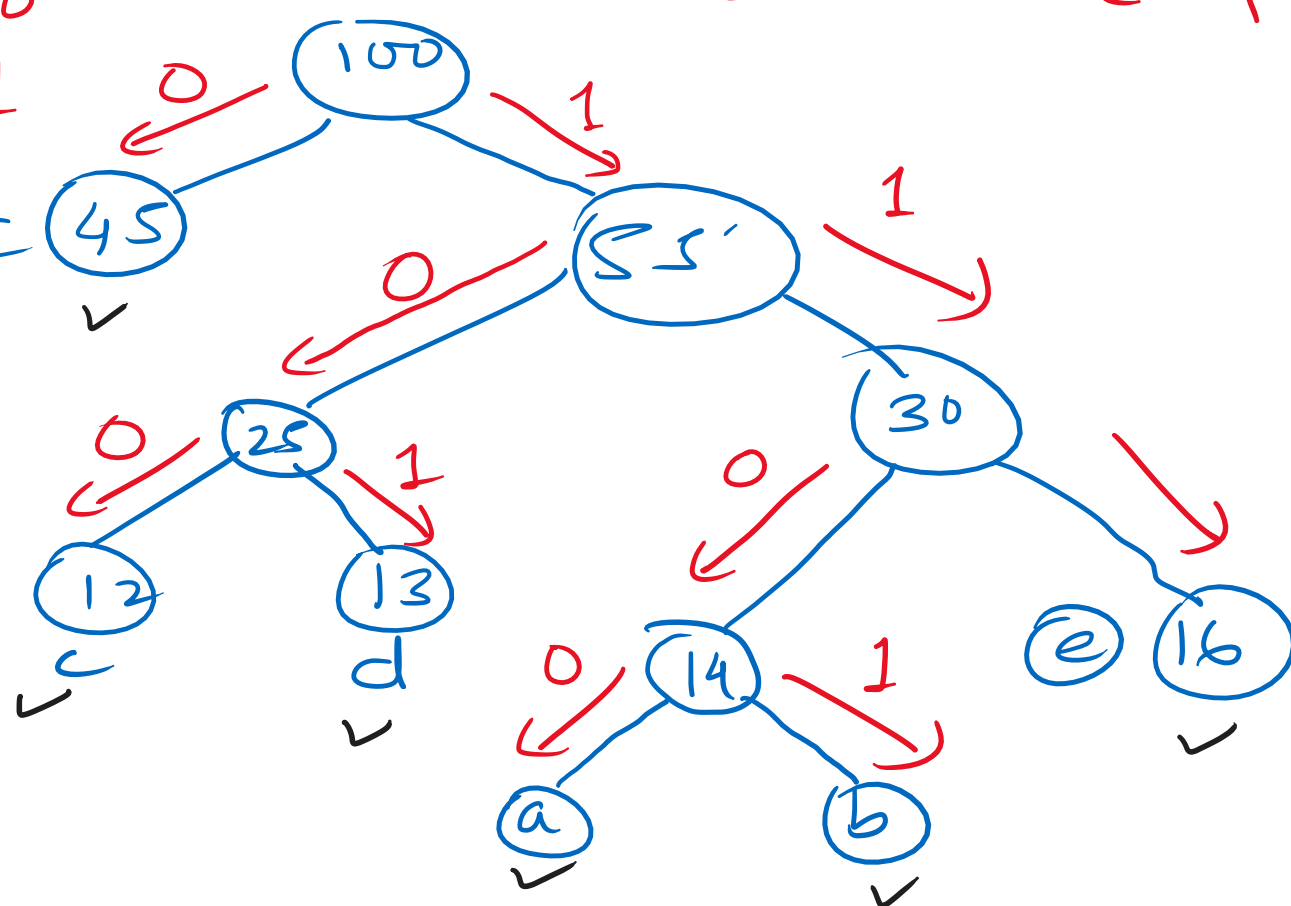
Huffman Encoding: $\rightarrow$

i/p $\rightarrow$ s = "abcdef"

freq = {5, 9, 12, 13, 16, 45}

f $\rightarrow$ 0
c $\rightarrow$ 100
d $\rightarrow$ 101
a $\rightarrow$ 1100
b $\rightarrow$ 1101
e $\rightarrow$ 111

Leaf Nodes



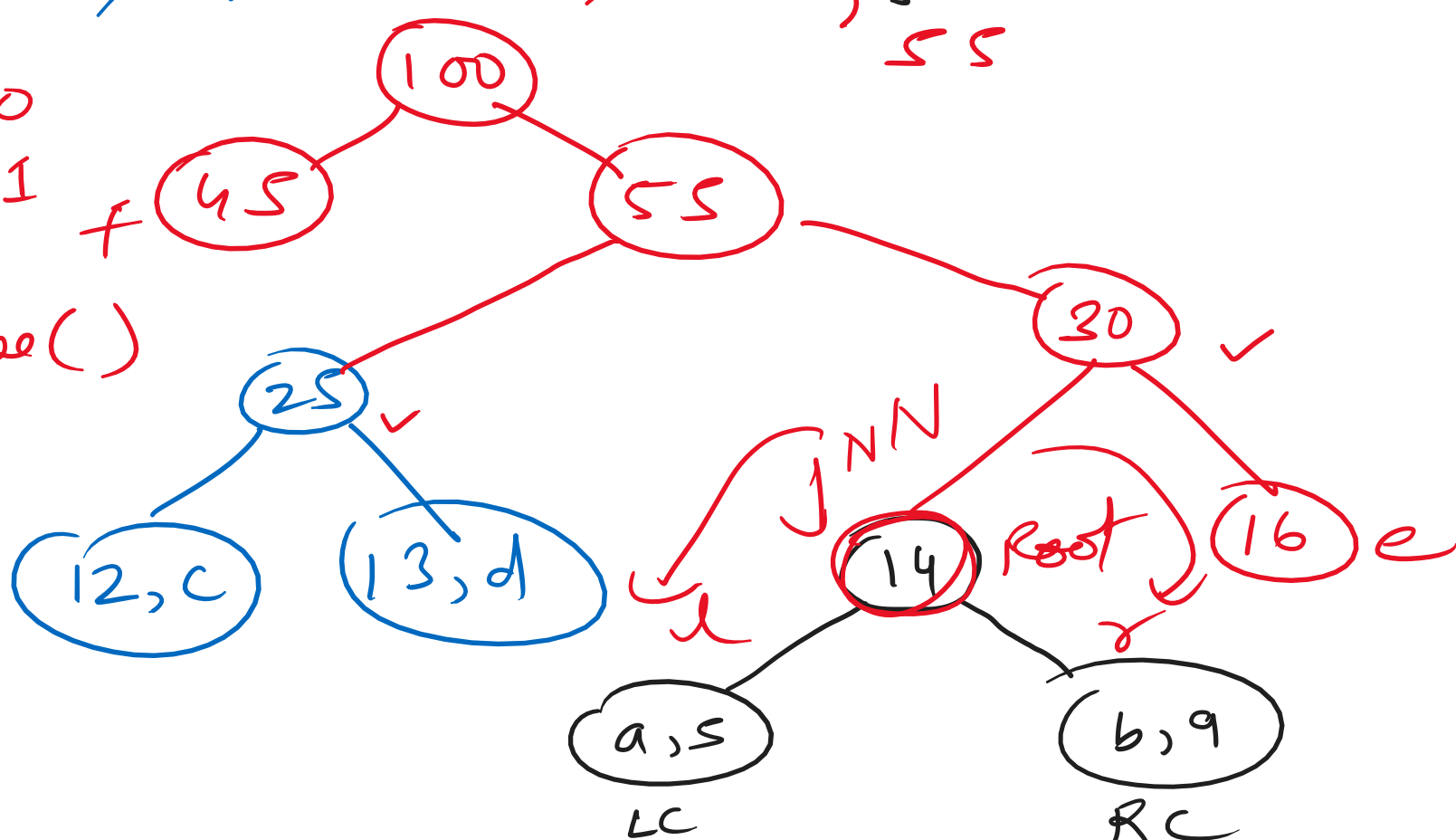
How to draw the tree? (min-heap)

$\{ \cancel{5}, \cancel{9}, \cancel{12}, \cancel{13}, \cancel{16}, 45 \}$ (pq)

L $\rightarrow$ 0

R $\rightarrow$ 1

traverse()



Searching & Sorting

DSA $\rightarrow$ Arrays, Strings, Linked List, Stack, Queues, Trees, Tries, Heaps, Graphs, Algos, DP, Greedy, Recursion, Back Tracking.

[Kadane Algo]

* Questions